

FilamentTracker Live Spool

Bill of Materials

Required and recommended hardware for building a FilamentTracker Live Spool appliance. The printable enclosure is modified from the SpoolBuddy enclosure by MartinNYHC and is used under the Creative Commons Attribution 4.0 International license.

Core components

Raspberry Pi 5 (4 GB or 8 GB)	Qty 1	Main controller
GPIO breakout board	Qty 1	Recommended for wiring and expansion
ELECROW 7-inch 1024x600 touchscreen	Qty 1	Primary user interface
PN532 NFC reader (I2C)	Qty 1	Reads/writes FilamentTracker NFC tags
NAU7802 load-cell amplifier	Qty 1	Preferred ADC; HX711 also supported
5 kg load cell	Qty 1	Weight sensing
microSD card (32 GB minimum)	Qty 1	64-256 GB recommended
USB-C power supply (5 V, 5 A)	Qty 1	Raspberry Pi power
NTAG215 NFC tags	As needed	Spool identification
M2.5/M3 hardware	As needed	Screws, inserts, standoffs, feet

Build notes

- A GPIO breakout board simplifies wiring and future maintenance.
- PN532 and NAU7802 intentionally share Raspberry Pi I2C bus 1.
- Verify each breakout board's voltage requirements before wiring.
- Load-cell wire colors are not universal; confirm the cell datasheet.
- Printable parts and assembly documents are under hardware/ in the repository.

Printable enclosure attribution

Original design: SpoolBuddy enclosure by MartinNYHC (Martin Ziegler).

Source: <https://makerworld.com/en/models/2296982-spoolbuddy>

License: Creative Commons Attribution 4.0 International (CC BY 4.0).

The Live Spool version changes branding, internal dimensions, touchscreen height, Pi 5 cooler and expansion clearance, rear airflow, and the electronics layout. See hardware/ATTRIBUTION.md for the complete attribution and change record.